

Satellite Imagery-Based Property Valuation

A Multimodal Regression Approach Using Tabular Data and Satellite Images

1. Introduction

Accurate residential property valuation is a core task in real-estate analytics. Traditional pricing models rely heavily on structured attributes such as square footage, number of bedrooms, construction quality, and location coordinates. While these variables capture measurable aspects of a property, they often fail to fully represent qualitative factors such as neighborhood aesthetics, surrounding greenery, road connectivity, and overall environmental context—commonly referred to as *curb appeal*.

This project explores whether **satellite imagery**, when combined with traditional tabular housing data, can enhance property valuation models. By integrating visual context with numerical features through a **multimodal regression pipeline**, we aim to assess both the quantitative and qualitative contribution of satellite images to price prediction.

While multimodal models were explored extensively, results indicate that structured tabular features remain the dominant predictors, with satellite imagery providing complementary interpretability rather than significant accuracy gains.

2. Dataset Description

2.1 Tabular Data

The base dataset is derived from the **King County House Sales dataset**, consisting of approximately **16,000 residential properties**. Each record represents a unique property sale with the following key attributes:

- **price** (target variable)
- Structural features:
bedrooms, bathrooms, sqft_living, sqft_lot, floors
- Quality indicators:
condition, grade, view, waterfront
- Temporal features:
yr_built, yr_renovated
- Spatial features:
lat, long
- Neighborhood context:
sqft_living15, sqft_lot15

The target variable **price** exhibits a right-skewed distribution and is log-transformed for modeling stability.

2.2 Visual Data (Satellite Imagery)

To capture environmental context, satellite images were programmatically fetched using the **Mapbox Static Images API**. For each property:

- The image is centered on the property's latitude and longitude.
- Zoom level and resolution are fixed to ensure consistency.
- Images are stored locally and linked to property IDs.

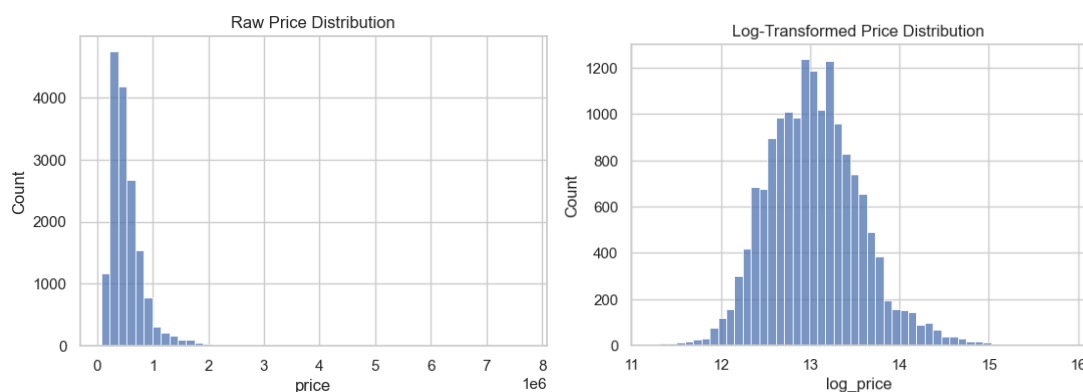
This approach allows the model to observe neighborhood layout, vegetation density, road patterns, and proximity to water bodies.

The dataset was split into training and validation sets using an 80–20 split.

3. Exploratory Data Analysis (EDA)

3.1 Price Distribution

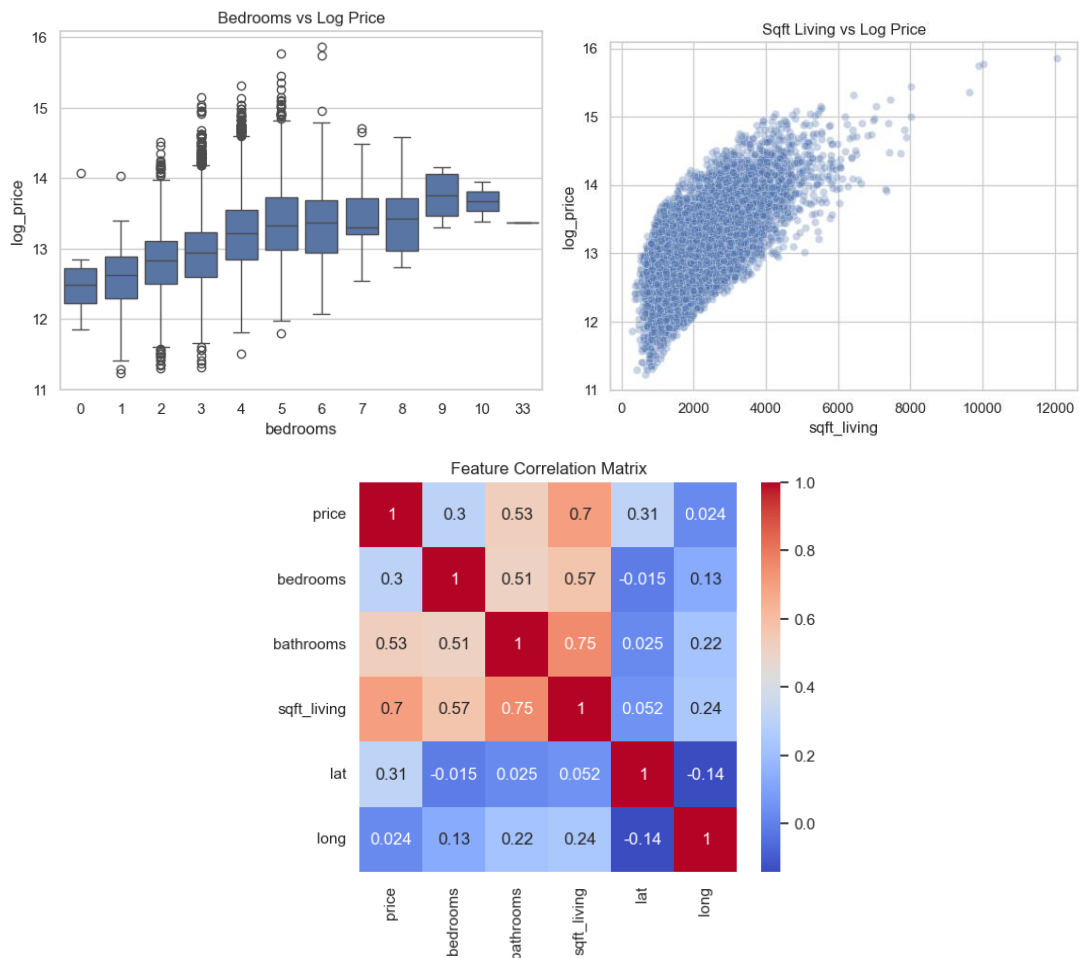
- Raw property prices show strong right skew.
- Log transformation results in a more symmetric distribution, improving regression performance.



3.2 Feature Relationships

Correlation analysis reveals:

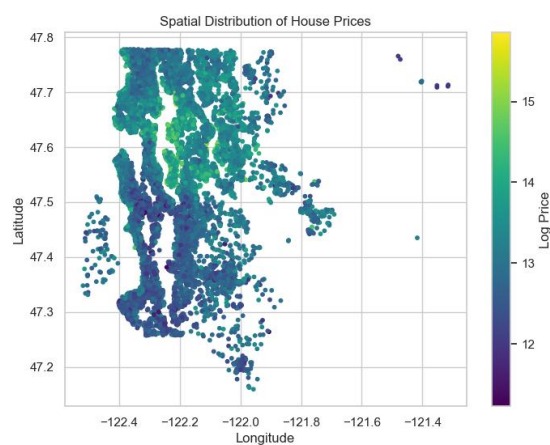
- Strong positive correlation between price and sqft_living, grade, and bathrooms.
- Moderate influence of geographic location (lat, long), indicating spatial clustering of prices.



3.3 Spatial Analysis

A latitude-longitude scatter plot colored by price shows:

- High-value properties concentrated in specific geographic regions.
- Clear spatial patterns that suggest neighborhood-level effects.



4. Image Feature Engineering

4.1 CNN-Based Feature Extraction

Satellite images are processed using a **pretrained ResNet-18 convolutional neural network**. Instead of training a CNN from scratch, transfer learning is applied:

- The final classification layer is removed.
- A **512-dimensional embedding** is extracted for each image.
- Embeddings are saved for reuse in downstream models.

This approach converts raw images into compact numerical representations capturing high-level visual patterns.

ResNet-18 was chosen due to its balance between representational power and computational efficiency.

4.2 Challenges

- Image embeddings are high-dimensional relative to dataset size.
- Visual features may be noisy compared to structured housing attributes.
- Careful fusion strategies are required to prevent overfitting.

5. Modeling Approaches

5.1 Tabular Baseline Model

A **Random Forest Regressor** is trained using only tabular features:

- Input: standardized numerical features
- Target: log-transformed price
- Evaluation metrics: RMSE and R^2

This model serves as a strong baseline due to the structured nature of the data.

5.2 Multimodal Fusion Models

Several fusion strategies are evaluated:

5.2.1 Late Fusion (Random Forest)

- Tabular features concatenated with image embeddings
- Trained using Random Forest regression

5.2.2 PCA-Based Fusion

- Image embeddings reduced using Principal Component Analysis
- Reduced embeddings concatenated with tabular features
- Trained using Random Forest

5.2.3 Neural Multimodal Model

- Separate neural subnetworks for tabular and image features
- Learned feature fusion via fully connected layers
- Trained using mean squared error loss

6. Results and Comparison

Model	RMSE (log-price) R ²	
Tabular Random Forest	0.179	0.884
Multimodal RF (PCA)	0.190	0.869
Neural Multimodal Model	0.251	0.772

Key Observations:

- The **tabular-only model outperforms multimodal models** quantitatively.
- Image features provide limited incremental predictive power.
- Neural models underperform due to dataset size and model complexity.
- The tabular Random Forest outperformed the best multimodal model by approximately 6% in RMSE.

7. Explainability with Grad-CAM

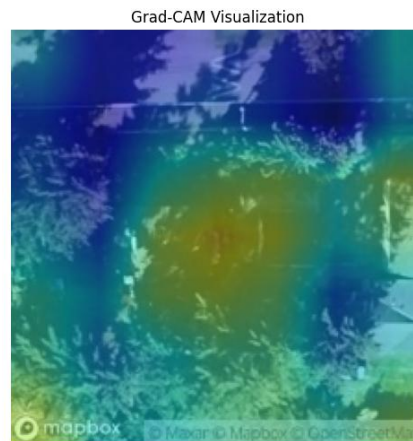
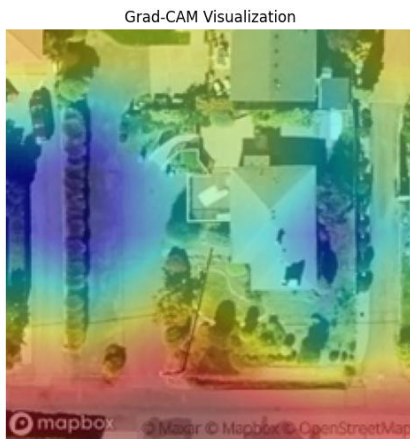
To interpret what the CNN learns from satellite imagery, **Grad-CAM (Gradient-weighted Class Activation Mapping)** is applied.

Grad-CAM was applied to the pretrained ResNet-18 backbone to interpret spatial attention patterns.

Findings:

- The CNN focuses on **neighborhood-level features**, not individual rooftops.
- High activation regions correspond to:
 - Green cover and vegetation
 - Residential density

- Road layout and connectivity



Interpretation:

Although satellite imagery does not significantly reduce prediction error, it captures **qualitative environmental cues** relevant to property desirability.

8. Discussion

The results highlight an important insight:

Strong structured features dominate price prediction in traditional housing datasets.

However, satellite imagery:

- Adds **interpretability**
- Provides **visual justification** for location-based effects
- Enhances understanding of environmental factors influencing valuation

This suggests that visual data may be more valuable for **explainability and decision support** than raw predictive accuracy alone.

9. Limitations and Future Work

Limitations:

- CNN embeddings are frozen (no fine-tuning)
- Image resolution is limited
- Dataset size restricts deep learning effectiveness

Future Improvements:

- Fine-tune CNNs on property-specific tasks
- Use higher-resolution or multi-scale imagery

- Incorporate temporal or street-level imagery
- Combine satellite data with socio-economic indicators

10. Conclusion

This project demonstrates a complete **multimodal machine learning pipeline** for property valuation. While traditional tabular models remain the strongest predictors, satellite imagery provides valuable spatial and environmental insights. The integration of explainable AI techniques such as Grad-CAM enhances transparency and supports more informed real-estate analytics.

In practice, such explainable multimodal systems can support real-estate analysts by providing qualitative context alongside quantitative price estimates.

References

1. King County House Sales Dataset – Kaggle
2. Mapbox Static Images API Documentation
3. Selvaraju et al., *Grad-CAM: Visual Explanations from Deep Networks*